Customer Linkage Application Overview

The Customer Linkage application allows you to create linkages in bulk between your DBS customers and Registry customers. You will be able to upload a file containing customers and create the DBS linkages. You will not be forced to create linkages as you will always have the option to remove the customers you uploaded without creating linkages. During this process, the Registry customer will not be updated with the DBS customer information uploaded and the customer record in your DBS will not be updated with Registry customer information. This process will only create a link (pointer) between the two customers to enable the user to know there is a link between the Registry customer and their DBS customer.

Glossary

DBS – Dealer Business System

EQUIP™ – This refers to the *EQUIP™* dealer business system version that the US and Canadian dealers use.

Registry (a.k.a. IKC, CKC) – John Deere's Customer Master Datasource, which is used by Service Center and many of John Deere's Sales Tools (i.e. Sales Center, JDQuote2, JDMint, JDSET, Configurator, Rewards, Warranty Portal, etc.).

Linkage – A link between a DBS customer and a Registry customer (like a pointer). The data between the two records can be different, but the record needs to be the same customer.

CSC (Common Search Component) – A tool used by many of the Sales Tools (i.e., JDQuote2, JDMint, JDSET, Configurator, Rewards, etc.) to search, add and edit customer data in Registry.

This component is embedded into the Sales Tools to ensure a consistent experience for the user. When a linkage is created between a DBS customer and a Registry customer then the DBS number will be displayed in CSC in the Membership column to indicate to the user that the customer in the search results has a linkage to a customer in their DBS. Linkages can only be seen by the dealership that created the linkage, so the user can be confident that the DBS number displaying belongs to their dealership.

Tight Match – Registry uses a tool to standardize names and addresses to compare the DBS customer record and the Registry customer record to see if they are the same customer. For example, if one record has "Bill" as the first name and the other record has "William" as the first name, then it would be treated as the same name. A tight match means the tool has determined that the 2 records are the same customer based on name and address.

Potential Match – If the customer you uploaded does not receive a tight match, then Registry's tool will look for customer records that are considered potential matches. A potential match is a record that is close to the customer uploaded, but the record does not exactly match it. It will take a person that knows the customer to determine if the potential match is the same customer. The phones, email and Tax ID uploaded in the DBS customer file will be used to find potential matching customer records in Registry. Only these countries can upload Tax ID: AR AU BO BR BZ CL CO CR DO EC GF GT GY HN HT JM M NI NZ PA PE PR PY SR SV TT UY VE

Company Use

Recommended/Proposed Approach

1. Initially start with a smaller list having 10 to 100 customer records. Possibly picking your top customers that you know you would like to link.
2. Clean the customer data in your DBS by removing any unneeded data and/or putting the data in the field that it belongs in.
   ** *EQUIP*™ Dealers: For tips on cleaning up your customer data in *EQUIP*™, please review the EQUIP Customer CleanUp document.
3. Extract the DBS customer data into the correct format and save as a csv file.
   **EQUIP™ Dealers: You can generate this file using the Contact Data Extract. For more details, please review the EQUIP Customer Linkage Contact Data Extract Document.
4. Upload the DBS customer list in the Customer Linkage application and create your linkages.
5. Communicate with your dealership personnel to ensure they understand what this new linkage is and that it will be seen in Sales Tools such as Sales Center, JDQuote2, JDMINT, JDSet, Rewards, Warranty Portal, etc., so they can take advantage of the linkage displaying in CSC's membership column.
6. Gather feedback from your dealership personnel.
7. Repeat the process with another set of customers that you would like to link.

Access Required

- To log into the application, your dealership needs to be set up to create linkages. If you receive the below message after attempting to log in, then you need to contact support to get your dealership set up to create linkages:

> ⚠ Your dealership is not set up to create linkages with Registry. If you would like to get your dealership set up to create linkages with Registry, please contact the Customer Linkage Feedback Team by sending an email to "CustomerLinkageFeedback@JohnDeere.com".

Also, you need to have a role assigned in DPA (Dealer Profile Administrator) for the Customer Linkage application. There are 2 DPA roles available:

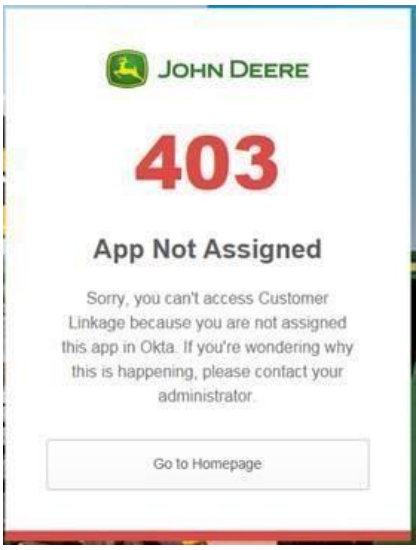
- Dealer Customer Maintainer – This role will be able to upload files plus has "Dealer Customer Reviewer" access.
- Dealer Customer Reviewer – This role will be able to create linkages for tight matches plus review customers and create linkages on the "Review" page.

If you receive any of the below messages after attempting to log in, then you do not have a role assigned in DPA for the Customer Linkage application:

Message 1:

> ⚠ You do not have a DPA role assigned to log into the Customer Linkage Tool. Please contact your Dealer Profile Administrator at your dealership to assign a role to log in to the Customer Linkage Tool.

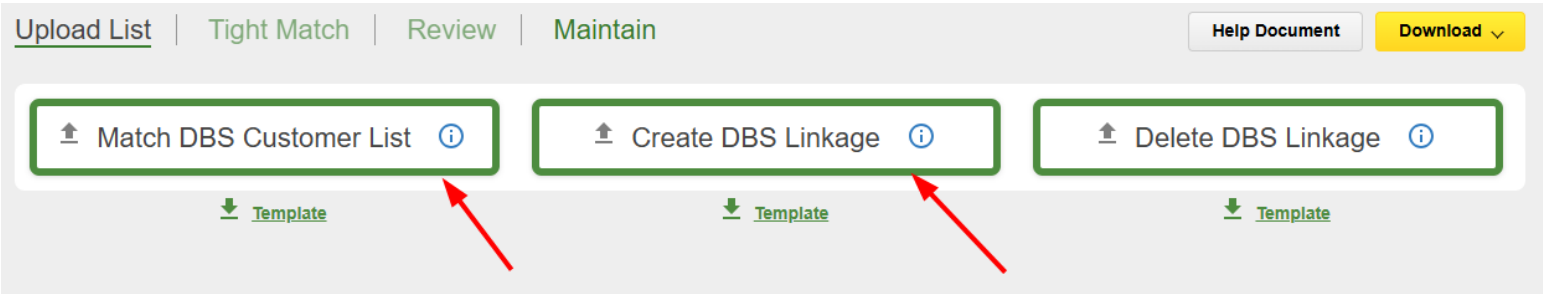
Message 2:



Creating DBS Linkages

There are two ways to create DBS linkages using the Customer Linkage Tool.

- **Match DBS Customer List:** You can download a list of customer records out of your DBS and upload it into the tool using this option. This will find matching customer records in John Deere and allow you to create the linkages between your DBS customer and the John Deere customer.
- **Create DBS Linkage:** If your dealership has stored the John Deere customer Entity ID in your DBS or CRM, you can upload the DBS Number and Entity ID to create the DBS Linkage using this option, which means you do not have to go through the customer data matching process. Contact ID is optional and should only be used if there is a need to link to a specific contact under a John Deere business customer record.

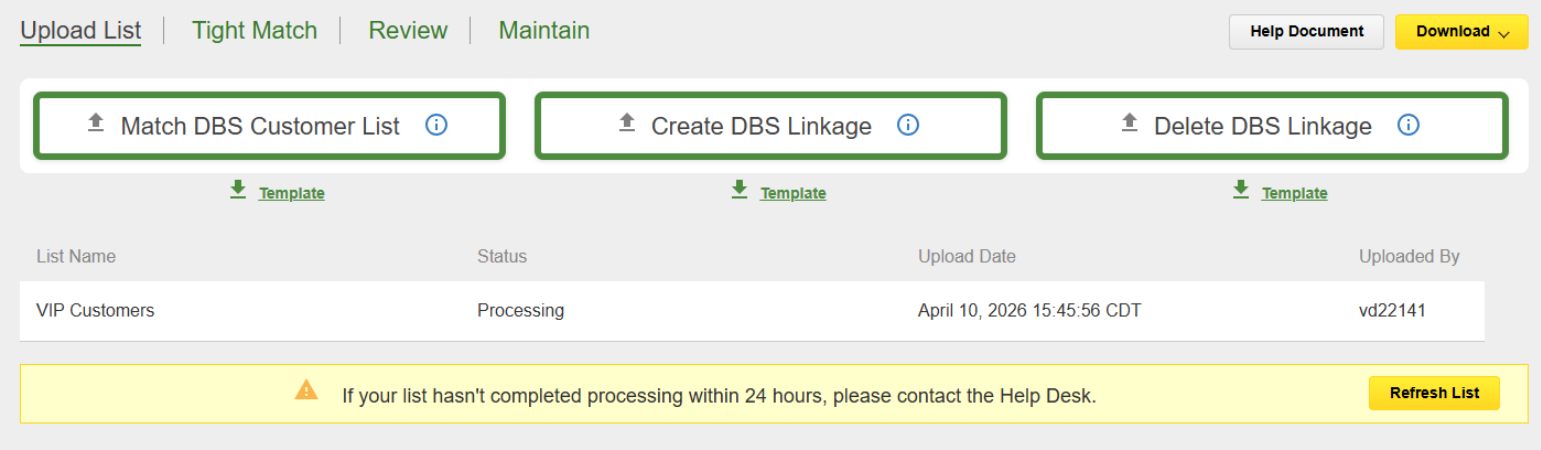


"Create DBS Linkage" Option

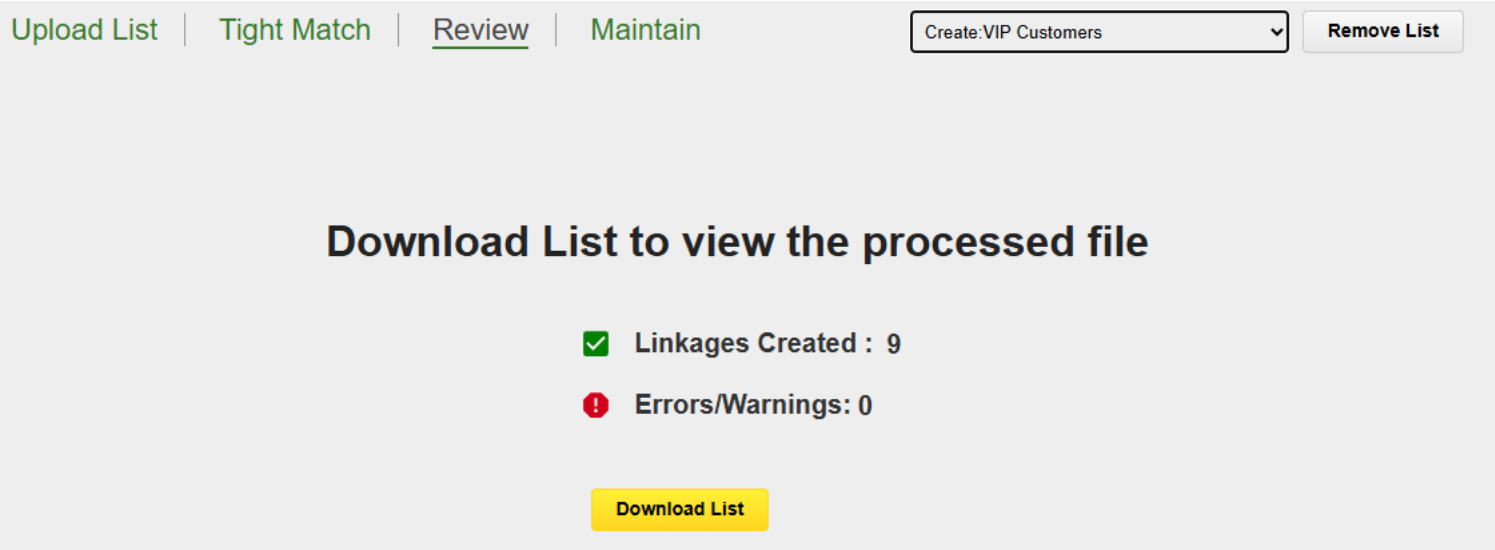
- Once logged into the Customer Linkage application, you will be on the "Upload List" page, where you will be able to Download the Linkage Report.
- If you have the appropriate DPA role, you will be able to upload a file and download templates.
    - Select your file by clicking on "Create DBS Linkage" and navigating to where your file is located.
    - OR drag and drop the file into the rectangle.
- Once your file has been selected, the first 20 rows will be displayed on a preview screen. Please carefully review all the columns and the information in your file to ensure they align. If

they do not align then you need to click “Cancel Upload,” fix your upload file, and re-upload. If they do align, then click “Create Linkages” button.

- After the file is uploaded, you should see the file name with the status of “Processing” on the “Upload List” tab.



- Processing time varies upon the number of records in the uploaded list. Once the file is done processing, you will receive an email. Click "Refresh List" to see if the processing is complete. If your file disappears from this screen, then it is done processing and you can go to the "Review" tab to download the results of the file.
- Once your file is done processing, go to the "Review" tab. This type of file will have "Create:" in front of the file name. Select the file from the dropdown menu to view and download the results of the file if needed.
- Once done with the file, the file can be removed from the dropdown by clicking the "Remove List" button.



Creating a File of DBS Customers to Upload using “Match DBS Customer List”:

Before creating linkages using the "Match DBS Customer List" option, a file containing customer information must be created. A template is provided on the “Upload List” page. This template will have the necessary columns needed to upload your customer data. If you do not have data for a column, then leave the data blank for that row.

> **EQUIP™ Dealers: You can generate this file using the Contact Data Extract. For more details, please review the EQUIP Customer Linkage Contact Data Extract Document.

*If the customer is an individual, then the Business Name columns should be left blank.
**If the customer is a business, then the First and Last Name columns may be blank if a business contact is unknown.

Address tips:

- Enter the physical address here if available; if not, enter the mailing address. Do not combine a physical and mailing address into one address.
- State codes must be capitalized character codes: IA, IL, FL, CA, MB, BC, NSW
- Country codes must be capitalized 2-character codes: US, CA, AU, NZ

Additional Customer Information tips:

- A list of valid values can be found in the dropdowns on the “Add Individual Page” in CSC for the following customer fields:
  - Prefix
  - Generation
- Linkages cannot be created to the customer records having spaces before or within the DBS number. It will reject the records and logs as an error file. It will create linkages for customer records having the DBS numbers with trailing spaces, but the spaces will be removed before linking.

File tips:

- The file must be saved in a .csv format. If the DBS customer data has special characters, please save the .csv file with UTF-8 format.
- The file needs to include column headings as the first row will not be processed plus it will help when you review the uploaded data to ensure the columns match up to what the application expects.
- A file should not be bigger than 6 MB (around 60,000 rows of customer information). If you have list exceeding this size than the file should be broken into smaller files. Each file will need to be uploaded separately.

**NOTE: If you have data in either of the business name fields and any of the prefix to suffix fields then Registry will never tight match to the business record. If you prefer a link on the business customer and not on the business contact before you upload the file into the tool you need to remove any business contact data from business records. If the customer record has a business name, then you will need to remove the data from the business contact fields, which are the following fields:

- Prefix

Company Use

- First Name
- Familiar Name
- Middle Name
- Last Name
- Generation
- Suffix

****EQUIP™** Dealers: If you wish to link your “Business with Contact” contacts directly to Business in Registry then when downloading the Customer Linkage Contact Data Extract, you can check the “Exclude Business Contact Name for Businesses with Contacts?” checkbox. This will remove the contact’s name (Prefix, First Name, Familiar Name, Middle Name, Middle Name, Last Name, Generation, Suffix) in the results returned.

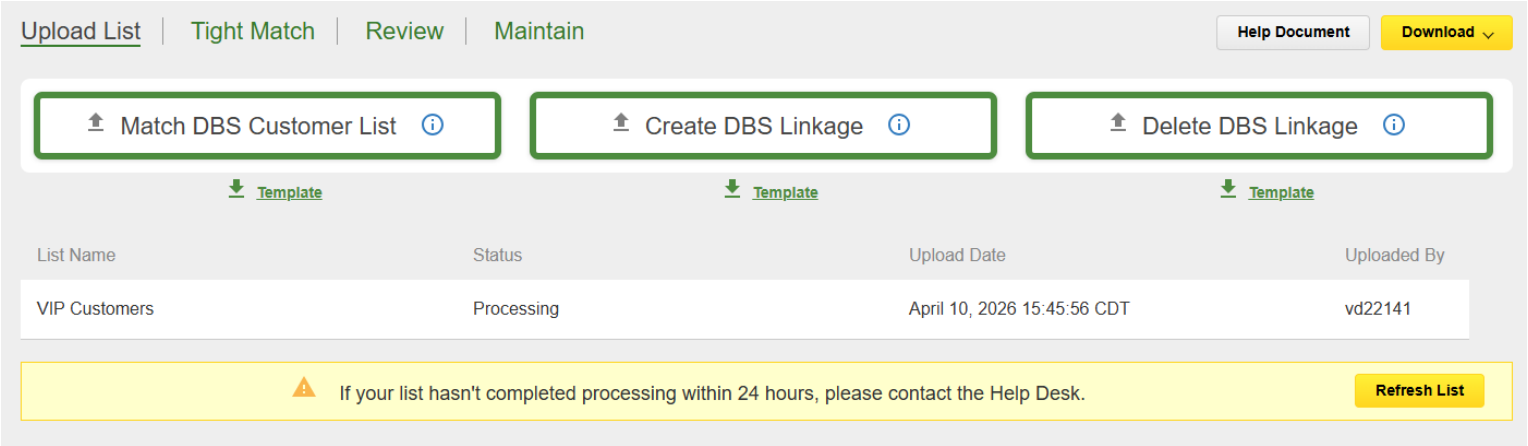
“Match DBS Customer List” Option

- Once logged into the Customer Linkage application, you will be on the “Upload List” page, where you will be able to Download the Linkage Report.
- If you have the appropriate DPA role, you will be able to upload a file and download templates. Select your file by clicking on “Match DBS Customer List” and navigating to where your file is located. (OR drag and drop the file into the rectangle.)
- Once your file has been selected, the first 20 rows will be displayed on a preview screen. Please carefully review all the columns and the information in your file to ensure they align. If they do not align then you need to click “Cancel Upload,” fix your upload file, and re-upload. If they do align, then click “Proceed.”

**NOTE: You can click anywhere inside the box (white area) to use your left, right, up, and down arrows to scroll horizontally or vertically.

Company Use

- After the file is uploaded, you should see the file name with the status of “Processing” on the “Upload List” page.



- Processing time varies upon the number of records in the uploaded list. Once the file is done processing, you will receive an email including these statistics:
    - Count of records in the file (not including the header row)
    - Count of duplicate records in the file
    - Count of records that already have a customer linkage established
    - Count of records that received a tight match
    - Count of records that received a potential match
    - Count of records with no match
    - Count of records with error.

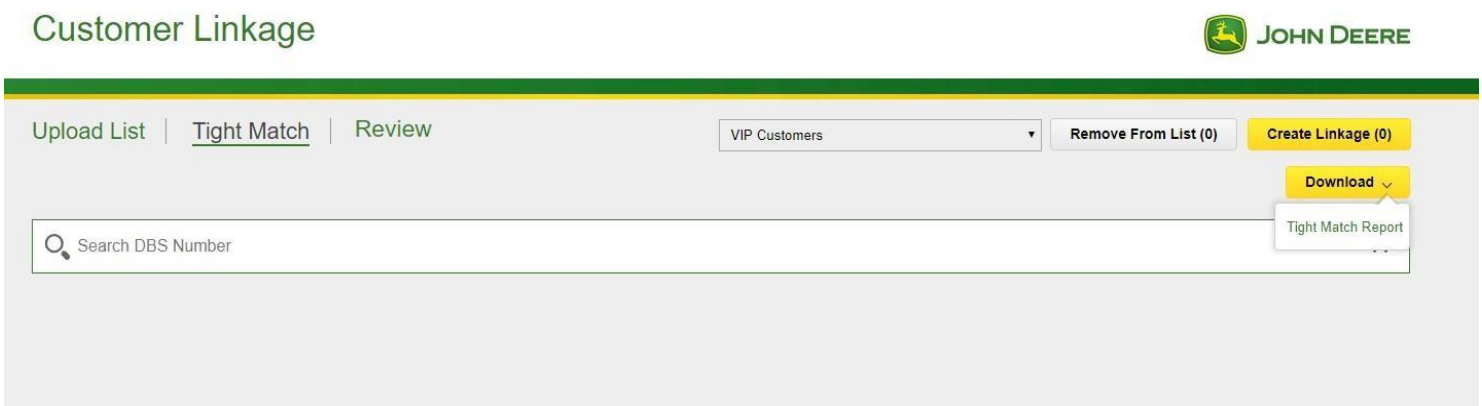
  **NOTE: On average, it takes one minute to process 500 records.

If any of your customer records received an error, then in the email some of the errors will be shown. If you need a full listing of your errors, please forward the email to 90ISCustomerLinkage@JohnDeere.com asking to receive all the errors. If you would like to re-upload the customers that failed, you will need to correct the records and upload a new file.
Now you are ready to review the matches and create the DBS linkages.

Creating Linkages for Tight Match Customers

Click on “Tight Match” page and select your file from the dropdown. If any of the customers in your file received a tight match with a Registry customer, then there will be a count on the “Remove From List” button and “Create Linkage” button on the “Tight Match” page.

Company Use

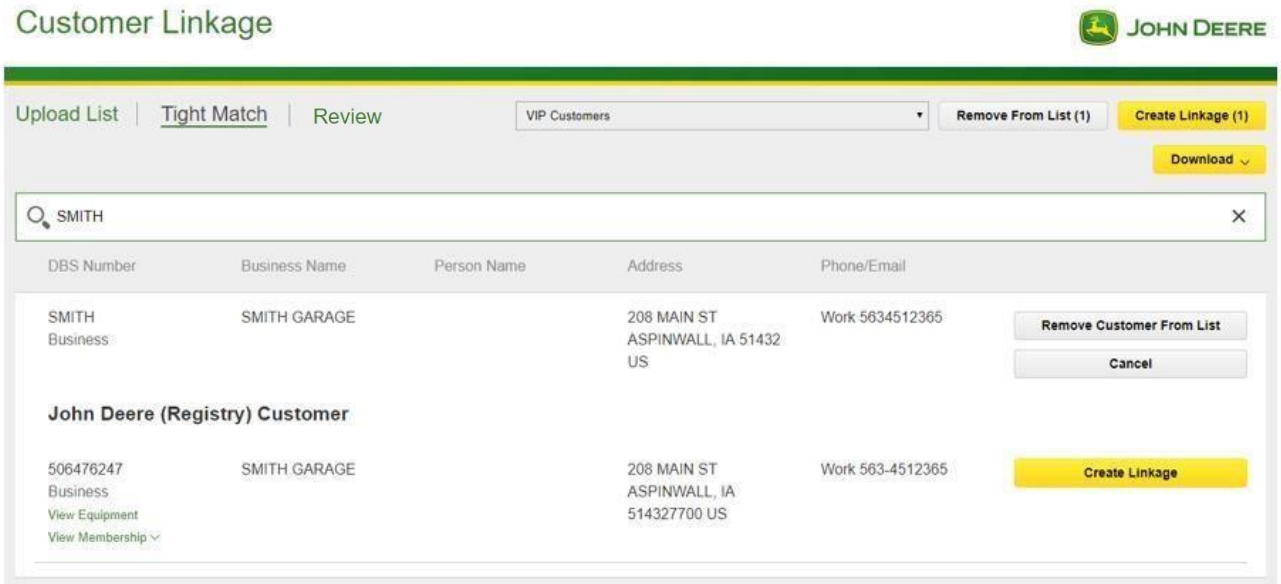


From this page you can:

- "Remove From List" – If clicked, this button will remove all the customers from the list that received a tight match, and no linkages will be created. You will only want to do this if you changed your mind about creating linkages for these customers.
- "Create Linkage" – If clicked, this button will create the linkages between the customers you uploaded and the Registry customers they tight matched to. Once the linkage is created, you will be able to see the DBS number displayed in CSC in the search results "Membership" column. If there are thousands of linkages to be created, this process may take overnight.
- "Download/Tight Match Report" – This button will allow you to download the DBS Customer records who received a Tight Match.

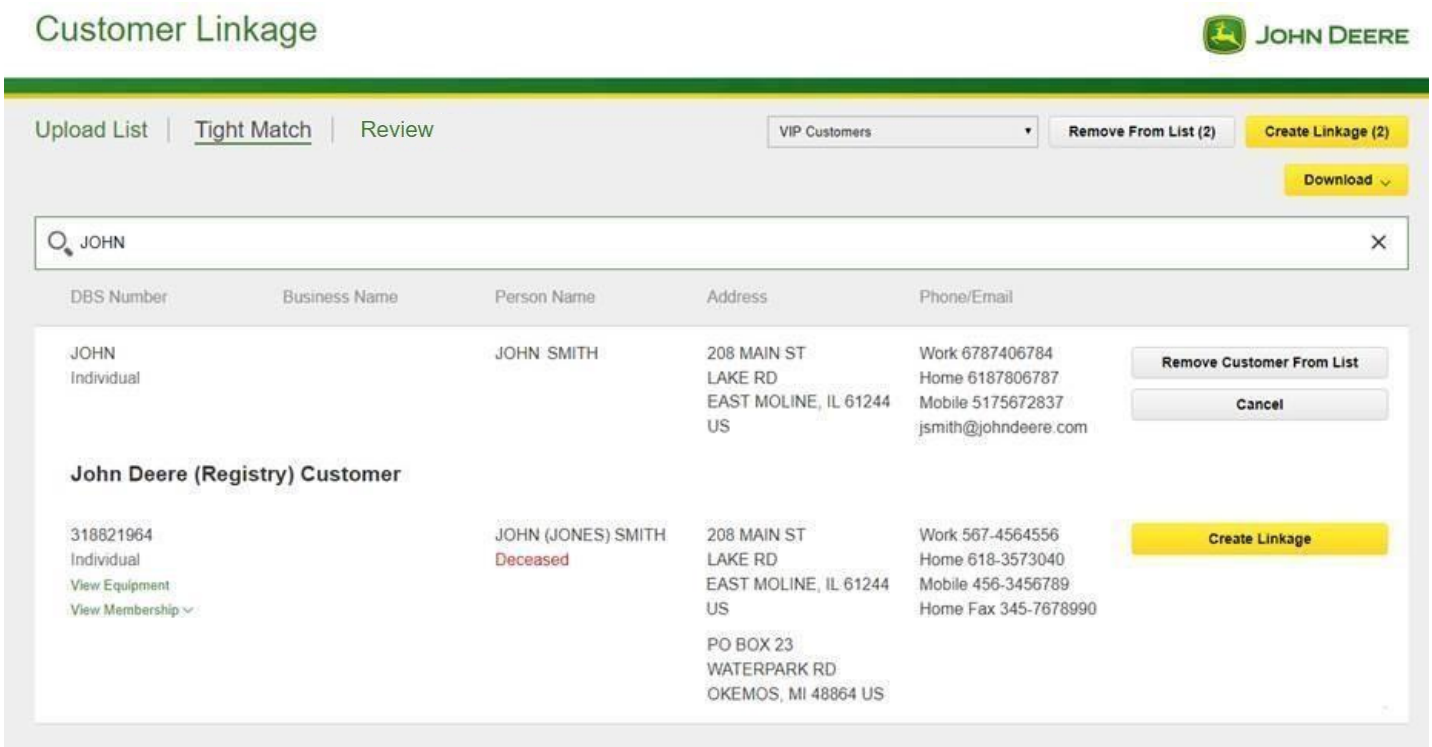
A	B	C
DBS CUSTOMER NUMBER	CUSTOMER ENTITY ID	CONTACT ID
SMITH	506476247	0

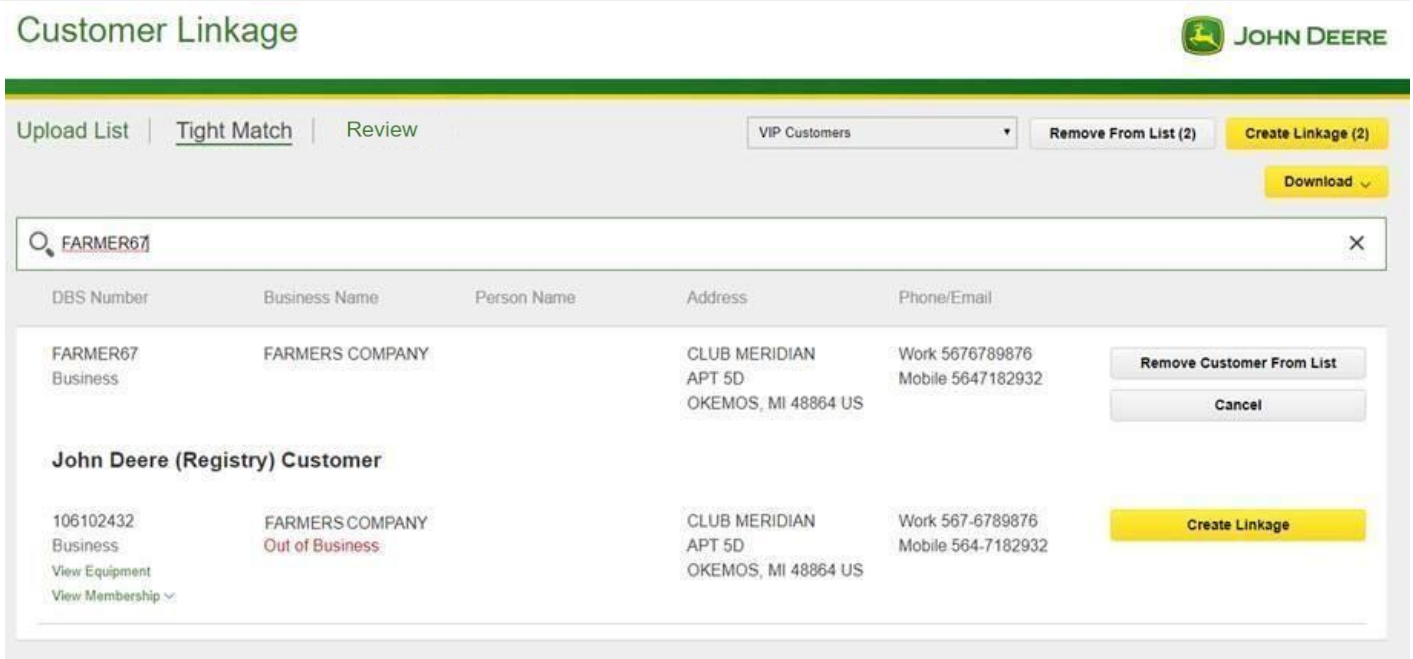
- Exact Search - Allows you to search using a DBS number to find a customer. If the customer received a tight match with the Registry customer, then the customers will be displayed on the "Tight Match" page.



Company Use

Note: On the "Tight Match" tab and "Review" tab, "Deceased" or "Out of Business" will display when the Registry customers are Deceased or Out of Business (see below).



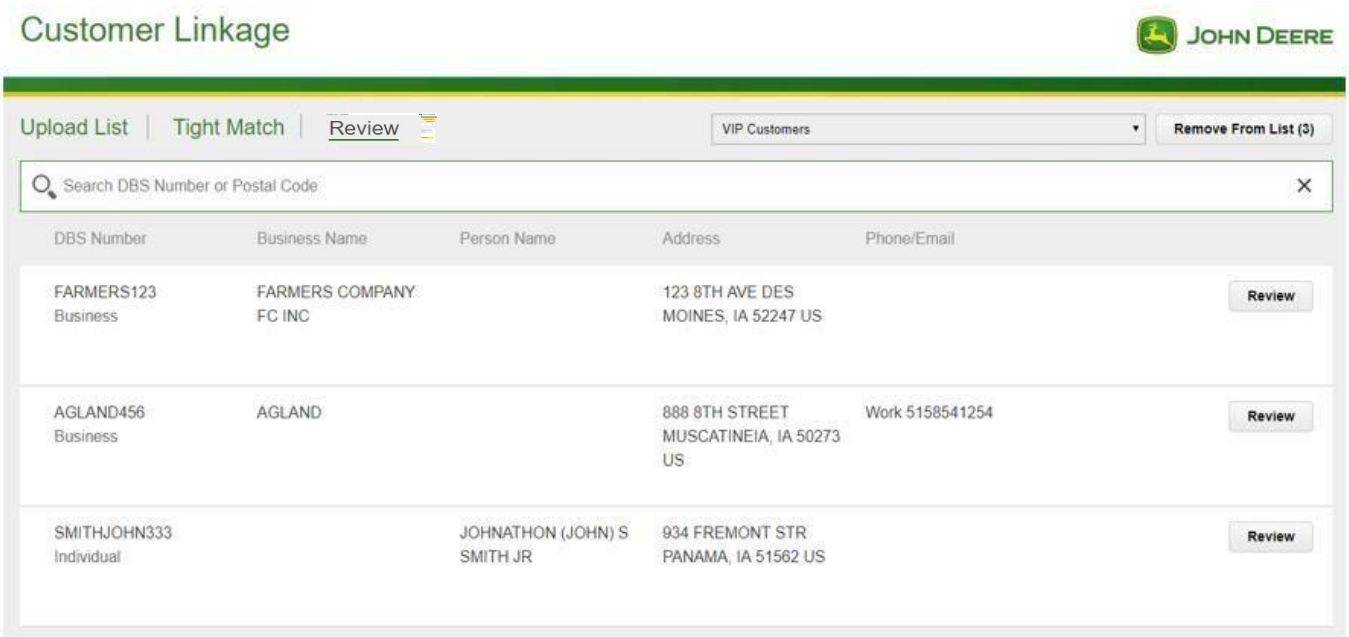


Review Customers that Did Not Receive a Tight Match

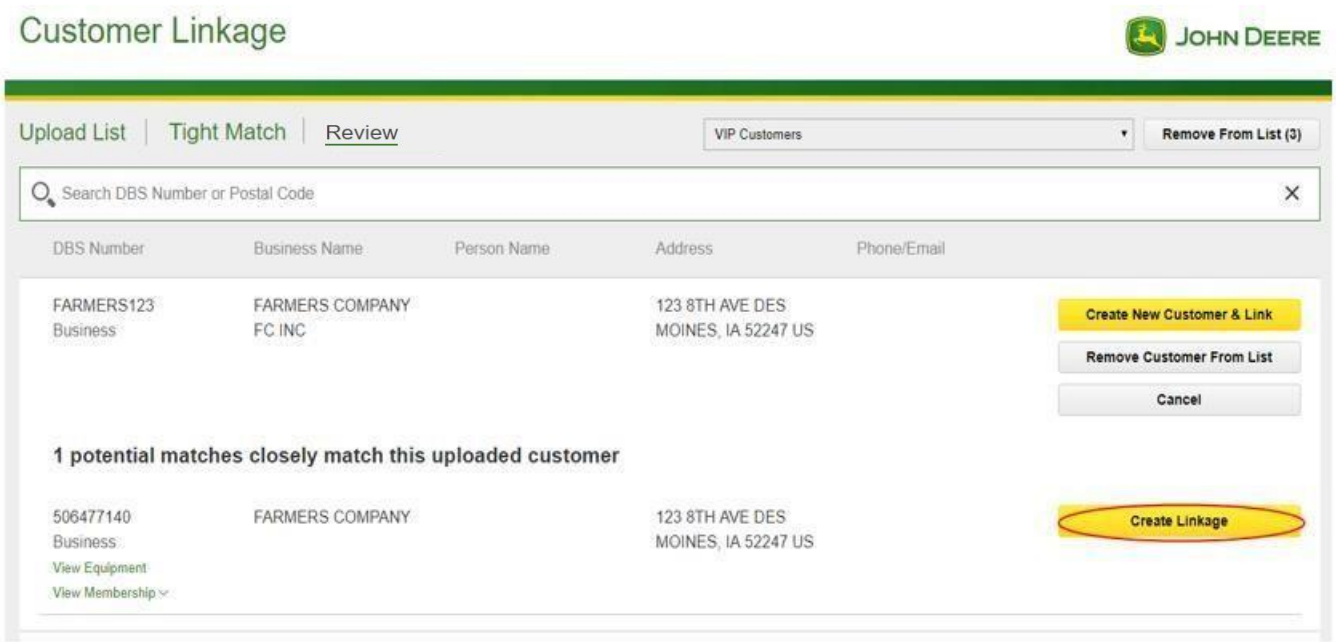
If any of your customers received a potential match with a Registry customer or did not receive a match, then these customers will display on the “Review” page. The customers that received a potential match will sort to the top of the list. You will be able to review each of these records to see if the customer you uploaded has any potential matches with Registry. If you do not want to review the customers on the "Review" page and create linkages, then you can remove all these customers from the list by clicking

Company Use

“Remove From List” button at the top of the “Review” page. This button will have the count of customers in your list in the file selected on this page.



- Navigate to the “Review” page and select the file you want to review from the dropdown. The application will display 30 customers at a time for review.
- If needed, you can filter your list using the search functionality by searching on the DBS Number or Postal Code this will be a “Starts With” search.
- Click the “Review” button next to the DBS customer you want to review. This will display the potential Registry records. There may be zero, one, or multiple. If any of the potential matches are the same customer, then click “Create Linkage.” This will create the DBS linkage to that Registry Customer and remove the DBS customer from the list.



Company Use

- If no potential matches were found or if none of the potential matches found are the same as your DBS customer, then you can add the customer into Registry by clicking on the “Create New Customer & Link.” The customer data you provided in the file will be pre-populated onto the CSC add customer screen. Fill out any additional information and add the customer. This will send the customer information to Registry. Once the customer is added successfully in Registry, the linkage will be created and the DBS customer will be removed from the list.

- If you do not want to create a linkage for this particular DBS customer you uploaded, you can click on “Remove Customer From List”, which will remove the DBS customer from the list and not create a linkage.

After creating the DBS linkage and you search for the customer in CSC, the membership will be displayed as shown below.

** *EQUIP*™ Dealers: Linkages created in Registry via the Customer Linkage Application will be sent to *EQUIP*™ overnight.

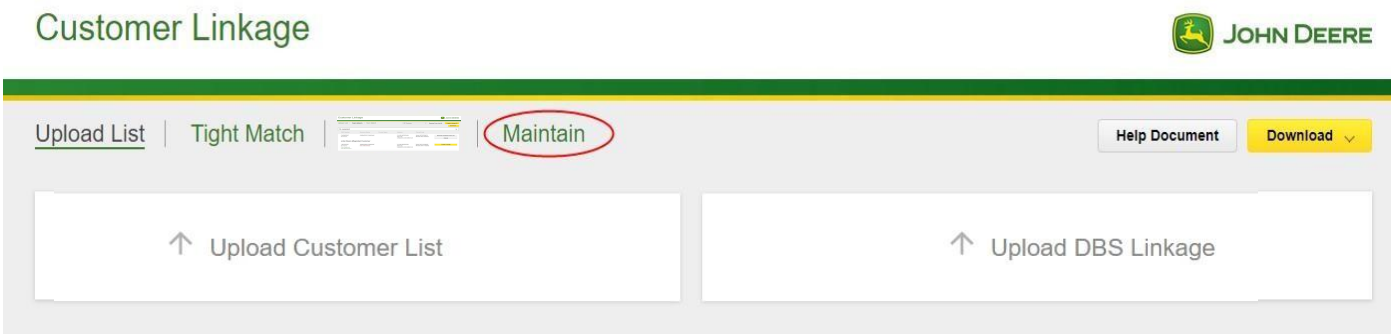
Remove Linkage Functionality

** *EQUIP*™ Dealers: Linkages can be removed through *EQUIP*™.

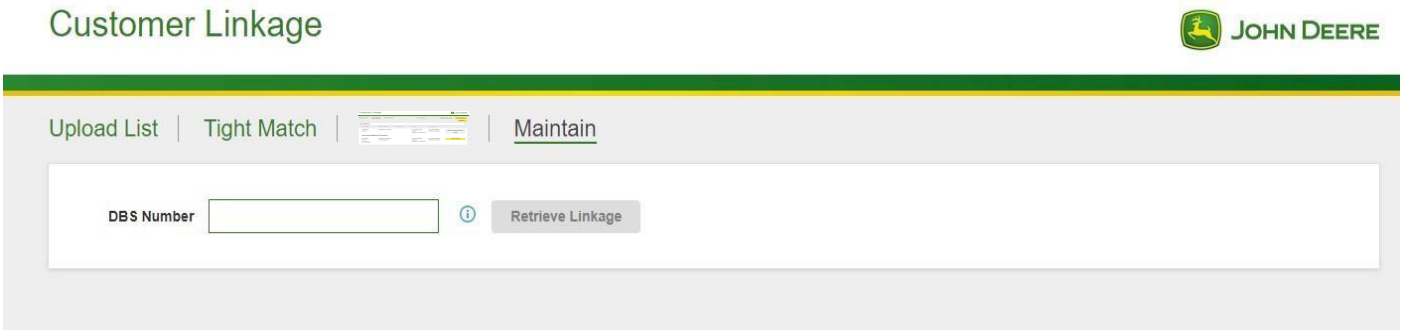
- Once logged into the Customer Linkage application, you will see the “Maintain” page.

Company Use

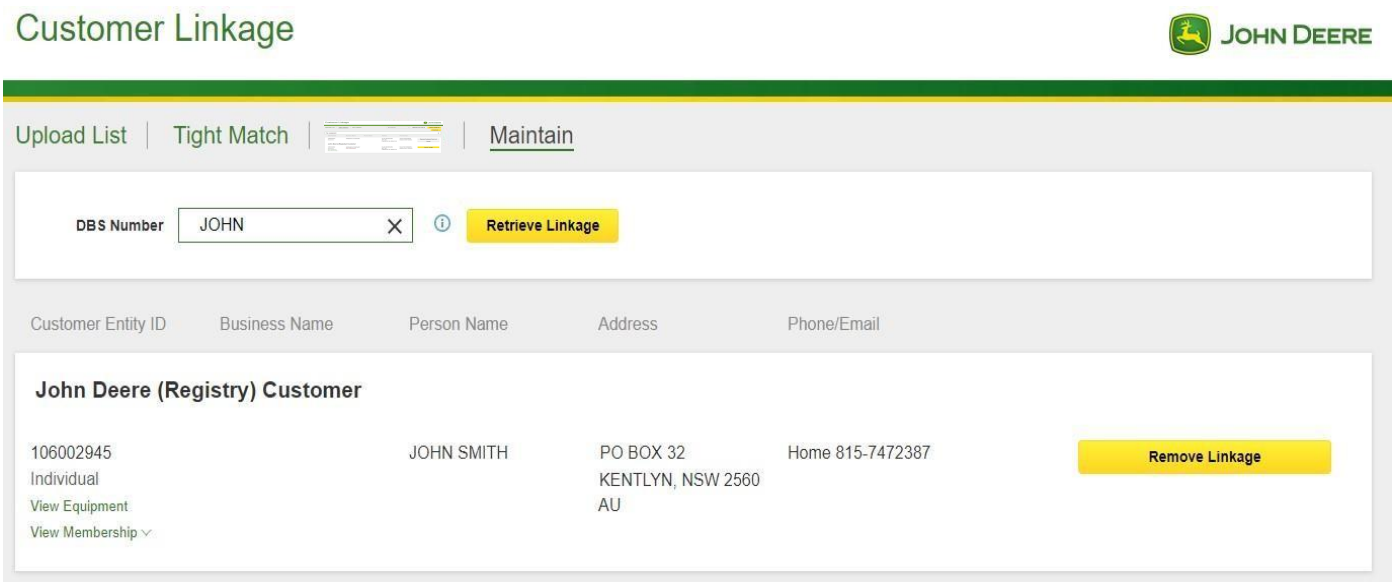
Note: Maintain page will not be available *EQUIP*™ dealers.



- Navigate to “Maintain” page, from this page you can search with DBS Number and retrieve the linkage if it exists.

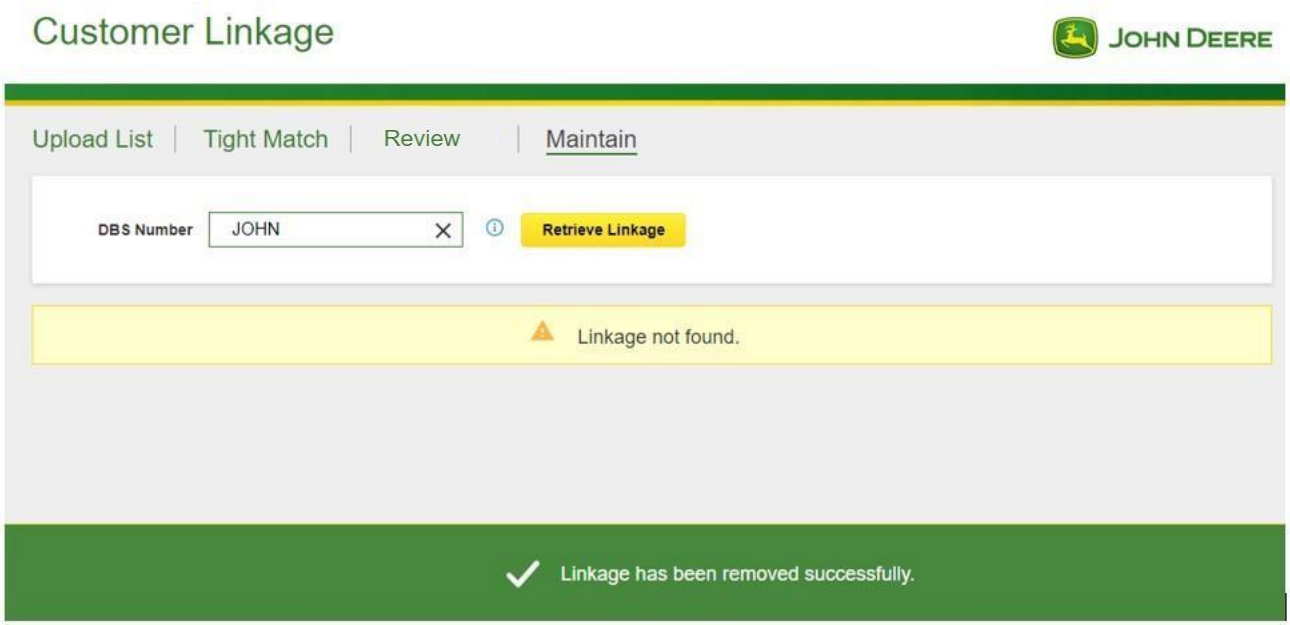


- Type in the DBS Number into the search box and click the “Retrieve Linkage” button. If a linkage exists, it will return the Registry Customer associated with that DBS Number and enable the “Remove Linkage” button.



- Click the “Remove Linkage” button. The linkage will be removed from Registry and displays a success message “Linkage has been removed successfully.” It will re-execute the search with the same DBS Number and returns a message “Linkage not found” because now the linkage does not exist.

Company Use



View Equipment

The “View Equipment” functionality will be available for Registry customers. This will retrieve all the equipment associated to this customer that is visible to your dealership in JDAim.

Note: JDAim access is required to retrieve the equipment. JDAim access is provided by the Dealer Profile Administrator at the dealership.

Potential Duplicate DBS Customer Listing

To identify the DBS Numbers that are linked to the same Registry Customer. Please review the Potential Duplicate DBS Customer Listing.

DBS Data Share (only available to US/CA *EQUIP*™ Dealers)

This is called the “DBS Data Share” because it allows you to share data between the two records as you choose. The DBS Data Share functionality is found within the Customer Search Component (CSC 2.0) that is used by John Deere Sales Tools and other applications. It is not available for JD Financial Customer records.

Before getting started, here are a few examples of data that SHOULD NOT be shared with Registry:

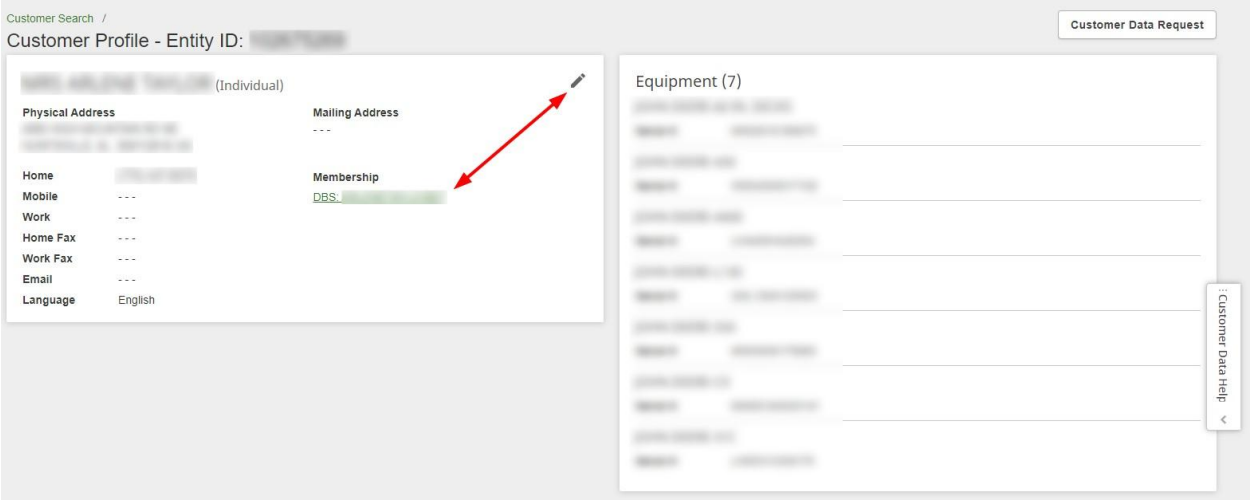
- In *EQUIP*™, there is “John & Mary” in the first name field. This data should not be shared with the Registry because in Registry every individual should have their own independent customer record.
- In *EQUIP*™, there is “C/O Bill” in the address line 1 field. This data should not be put into the Registry because it is not a valid part of the address.

Company Use

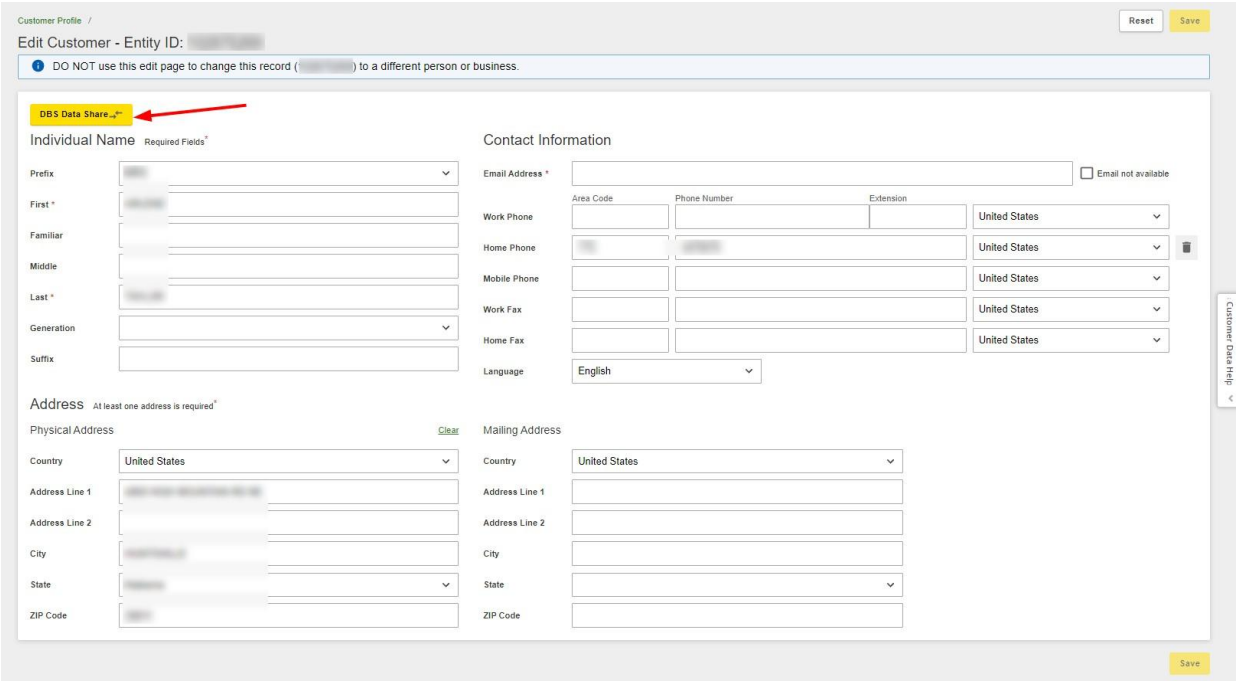
Note: This type of data can remain within EQUIP™ because that is your Dealer Business System customer record. So, you can keep the record as you choose. It is acceptable that there are differences between the two customer records if the customer records really are for the same customer.

How to use the DBS Data Share functionality:

- Within the Search Component (CSC 2.0) that is used by John Deere Sales Tools and John Deere applications.
- Search for any customer in CSC 2.0 per your normal process
- Select the customer to view that customers "Customer Profile" information.
- If the Registry customer has a DBS linkage, then the *EQUIP*™ Contact Code (preceded by "DBS:") will be displayed under "Membership".
- To edit the customer record, the user can click on the pencil icon.



- On the "Edit Customer" page the user will see the "DBS Data Share" button.
- The user can choose to only edit the Registry customer record **OR** click on the "DBS Data Share" button to edit the Registry customer and *EQUIP™* customer record at the same time.



Company Use

Access to this Data Share functionality is controlled by the Customer Linkage DPA roles (see snapshot below). If a user has one of these DPA roles, they can access this functionality.

Customer Linkage

- [x] Dealer Customer Maintainer
- [ ] Dealer Customer Reviewer

Frequently asked questions (FAQ's)

Q: What are the best practices for managing customer data?

A: The Customer Data Management Job Aid will provide best practices and tips on how everyone can contribute to high-quality customer data.

Q. What happens if I request to merge customers in CSC or JDAim?
A. If your merge request is fulfilled than the linkages established on the losing customer record will move over to the surviving customer record. For more details, please review the Customer Merge Request Process.

Q. What happens if I delete a business contact in CSC?
A. This contact will be deleted. The name, email and phones will be deleted. Any business related data (like DBS linkages) will be moved to the business.
** *EQUIP*™ Dealers: When customer merges or deletes happen the linkages will be sent to *EQUIP*™ overnight, so it will be reflected in *EQUIP*™ the day after Registry completes the merge/deletion.

Q. Is there a training regarding Customer Linkage and *EQUIP*™ Customer Data Clean Up?
A. Yes. Link to training video: https://p.widencdn.net/04rxtm/CustLinkageEQUIPCustDataCleanUp

Q: ** *EQUIP*™ Dealers: How to set up *EQUIP*™ to integrate (search customers and create/remove links) with Registry?

A. For more details, please review the KB0111655 EQUIP Set Up for Integration with Registry and Usage

Q. How many Registry customer types are there?
A. There are two customer types, Business and Individual. A Business can have zero, one or multiple contacts.
Business and Individual records will have an Entity ID, which will be displayed in CSC, JDAim and other applications.
Business Contact will have a Contact ID.
From the below example, the Business is having two Business Contacts.

Company Use

Registry Customer Types	Business Name	Person Name	Entity ID	Contact ID
Business	SMITH GARAGE		501065658	
Business with Contact	SMITH GARAGE	MICKEY SMITH	501065658	105263458
Business with Contact	SMITH GARAGE	JOHN SMITH	501065658	105263459
Individual		JACK KEAN	587894325	

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